Supplementary Material
Executable Scientific Contracts for Verifiable LLM and Agent Research

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1 Complete contract inventory

The inventory records the source, role, original classification, eligible data, and evidence for each of the 25 checks.

The τ-bench set contains two substantive checks, two checks of agreement between stored data fields, and one summary of preceding results. The reward check compares saved top-level and nested reward fields on 1,313 of 1,320 trajectories.

Original classifications. 15 conformant, seven discrepant, one inactive, and two convention sensitivities.

Table 1: The 25 original checks, with source locations, roles, recorded classifications, and eligible data counts.

Check	Source	Role	Original classification	Eligible data	Evidence
APRM-TEMP-01	compute_prm_target.py:55	scientific assertion	discrepant	24 controlled state/path observations. actual pinned callers and source variants	Original: history/submitted/results/reproduction/audits/agentprm.json Current: study/camera_ready/results/agentprm/challenge_results_v2.json
APRM-TEMP-02	compute_prm_target.py:55	consistency control	conformant	510 controlled state observations at gamma=1. actual source outputs in state sweep	Original: history/submitted/results/reproduction/audits/agentprm.json Current: study/camera_ready/results/agentprm/state_targets_v1.csv

Check	Source	Role	Original classification	Eligible data	Evidence
APRM-TEMP-03	compute_prm_target.py:55; compute_prm_preference_target.py:55	scientific assertion	discrepant	254 positive terminal-return state observations at released gamma. actual source outputs	Original: `history/submitted/results/reproduction/audits/agentprm.json` Current: `study/camera_ready/results/agentprm/state_targets_v1.csv`
APRM-SCHEMA-01	rollout_alfworld.py:66	scientific assertion	discrepant	6 controlled query-history entries. actual rollout AST extraction with recording agent/environment fixture	Original: `history/submitted/results/reproduction/audits/agentprm.json` Current: `study/camera_ready/results/agentprm/case_analysis_v1.json`
APRM-METRIC-01	compute_prm_target.py:55	superseded illustrative fixture	discrepant	0 upstream-valid instances of historical varying-outcome fixture. historical fixture superseded by valid constant-terminal trajectory and shared-prefix analyses	Original: `history/submitted/results/reproduction/audits/agentprm.json` Current: `study/camera_ready/results/agentprm/mean_scope_companion_v2.json`
CBEN-LOG-01	released Qwen3.5-9B slice; test_docker_runner_loop.py:77	scientific assertion	discrepant	99 released records. evaluated	Original: `history/submitted/results/reproduction/audits/contractbench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`

Check	Source	Role	Original classification	Eligible data	Evidence
CBEN-LOG-02	released Qwen3.5-9B slice	scientific assertion	conformant	Not evaluated: 0 eligible successful episodes.	Original: `history/submitted/results/reproduction/audits/contractbench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`
CBEN-SCHEMA-01	released Qwen3.5-9B slice	scientific assertion	conformant	99 released records. evaluated	Original: `history/submitted/results/reproduction/audits/contractbench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`
CBEN-METRIC-01	released Qwen3.5-9B slice; reported baseline	scientific assertion	discrepant	99 released records. evaluated	Original: `history/submitted/results/reproduction/audits/contractbench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`
CBEN-METRIC-02	released Qwen3.5-9B slice	consistency control	conformant	99 released records. evaluated	Original: `history/submitted/results/reproduction/audits/contractbench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`

Check	Source	Role	Original classification	Eligible data	Evidence
TACE-COORD-01	sft_dataset_wo_apply_template.py:170; sft_dataset_wo_apply_template.py:184; ToolACE support summary.json	scientific assertion	discrepant	56 controlled length-grid fixtures. actual unchanged loader and executable repair	Original: `history/submitted/results/reproduction/audits/toolace.json` Current: `study/camera_ready/results/toolace/challenge_results_verified_v2.json`
TACE-COORD-02	ToolACE support summary.json	activation check	conformant	9394 official nonempty tool-call targets. all official converted records evaluated, including separately identified EOS-only responses	Original: `history/submitted/results/reproduction/audits/toolace.json` Current: `study/camera_ready/results/toolace/case_analysis_v1.json`
TACE-SCHEMA-01	ToolACE support summary.json	scientific assertion	conformant	9394 official nonempty parsed targets. row-equivalent official conversion and parsed-name presence checks	Original: `history/submitted/results/reproduction/audits/toolace.json` Current: `study/camera_ready/results/toolace/case_analysis_v1.json`
TACE-MASK-01	sft_dataset_wo_apply_template.py:184	consistency control	conformant	9964 official converted examples. actual loader masks and trainer shift against position oracle	Original: `history/submitted/results/reproduction/audits/toolace.json` Current: `study/camera_ready/results/toolace/case_analysis_v1.json`

Check	Source	Role	Original classification	Eligible data	Evidence
TACE-SUPPORT-01	ToolACE support summary.json	activation check	inactive	9964 official converted examples. full official 8192 census	Original: `history/submitted/results/reproduction/audits/toolace.json` Current: `study/camera_ready/results/toolace/case_analysis_v1.json`
TBEN-SCHEMA-01	historical trajectories (two domains)	scientific assertion	conformant	5920 declared task actions. evaluated	Original: `history/submitted/results/reproduction/audits/taubench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`
TBEN-LOG-01	historical trajectories (two domains)	scientific assertion	conformant	1320 released records. evaluated	Original: `history/submitted/results/reproduction/audits/taubench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`
TBEN-METRIC-01	historical trajectories (two domains)	consistency control	conformant	1313 released records. evaluated	Original: `history/submitted/results/reproduction/audits/taubench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`

Check	Source	Role	Original classification	Eligible data	Evidence
TBEN-METRIC-02	historical trajectories (two domains)	consistency control	conformant	1320 released records. evaluated	Original: `history/submitted/results/reproduction/audits/taubench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`
TBEN-NEG-01	historical trajectories (two domains)	derived summary	conformant	4 preceding checks. evaluated	Original: `history/submitted/results/reproduction/audits/taubench.json` Current: `study/camera_ready/results/remaining_evidence_v3.json`
AABS-PAIR-01	README.md:23 (263 pairs; one should-act and one should-abstain task); README.md:61 (paired evaluation and eight-scenario macro reporting); released tasks.jsonl at locked dataset revision	scientific assertion	conformant	263 released act/abstain pairs. complete released-data census	Original: `history/submitted/results/agentabstain_prospective_v2.json` Current: `study/camera_ready/results/agentabstain/case_analysis_v1.json`
AABS-PAIR-02	AgentAbstain paper Appendix B.6, Eq. (3) (pair conjunction); README.md:61 (paired evaluation); eval/statistics/analysis.py:258-260; four-cell exact fixture	scientific assertion	conformant	4 controlled Boolean pairs. actual evaluator and four-cell complete executable example	Original: `history/submitted/results/agentabstain_prospective_v2.json` Current: `study/camera_ready/results/agentabstain/challenge_results_v2.json`

Check	Source	Role	Original classification	Eligible data	Evidence
AABS-CAR-01	AgentAbstain paper Appendix B.6, Eq. (4) (conditional definition); eval/statistics/analysis.py:231-243 (explicit zero fallback); conditioned-denominator exact fixture	convention sensitivity	convention sensitivity	2 conditioned pairs in nonempty controlled fixture. actual conditional metric plus separately guarded zero-denominator convention	Original: `history/submitted/results/agentabstain_prospective_v2.json` Current: `study/camera_ready/results/agentabstain/challenge_results_v2.json`
AABS-COMMIT-01	AgentAbstain paper Section 2.2 and Appendix B.6; README.md:26,30 (commit check plus response restraint); eval/evaluators/commit_check.py:12-15,44-49; eval/statistics/analysis.py:142-165; commit-by-response Cartesian fixture	scientific assertion	conformant	4 controlled commit/response Boolean cases. actual aggregate metric and success-filter evaluator extraction	Original: `history/submitted/results/agentabstain_prospective_v2.json` Current: `study/camera_ready/results/agentabstain/case_analysis_v1.json`
AABS-MACRO-01	AgentAbstain paper Section 3.1 and Figure 3 (macro over eight scenarios); README.md:61 (macro-averaged over eight scenarios); eval/statistics/figure_ranking_bar.py:103-120 (equal action-subtype weights inside scenario); unequal-action-type exact fixture	convention sensitivity	convention sensitivity	263 released pairs defining subtype weights. actual final ranking computation and exact composition bound	Original: `history/submitted/results/agentabstain_prospective_v2.json` Current: `study/camera_ready/results/agentabstain/case_analysis_v1.json`

2 Implementation cases

Table 2: Decisions on all 60 implementation cases.

Case	Expected	Field/type	Original adapters	Native tests	Behavioral contract	Eligible
APRM-DEV-TEMP-ORIGINAL-V2 terminal discount	fault	pass	fail	unavailable	fail	24
APRM-DEV-TEMP-REPAIR-V2 terminal discount	conformant	pass	fail	unavailable	pass	24
APRM-DEV-HIST-ORIGINAL-V2 query history action	fault	pass	fail	unavailable	fail	6
APRM-DEV-HIST-REPAIR-V2 query history action	conformant	pass	pass	unavailable	pass	6

Case	Expected	Field/type	Original adapters	Native tests	Behavioral contract	Eligible
APRM-LOCK-TEMP-OFFSET-V2 terminal discount	fault	pass	fail	unavailable	fail	24
APRM-LOCK-HIST-STALE-V2 query history action	fault	pass	pass	unavailable	fail	6
APRM-LOCK-TEMP-BENIGN-V2 terminal discount	conformant	pass	fail	unavailable	pass	24
APRM-LOCK-HIST-BENIGN-V2 query history action	conformant	pass	pass	unavailable	pass	6
APRM-LOCK-MISSING-V2 terminal discount	not evaluated	not evaluated	not evaluated	unavailable	not evaluated	0
APRM-LOCK-INELIGIBLE-V2 terminal discount	not evaluated	not evaluated	not evaluated	unavailable	not evaluated	0
TACE-ORIGINAL uncropped mask origin	fault	pass	fail	unavailable	fail	56
TACE-REPAIR uncropped mask origin	conformant	pass	fail	unavailable	pass	56
TACE-CROP-SIGN crop offset sign	fault	pass	fail	unavailable	fail	56
TACE-FIRST-RESPONSE first response boundary	fault	pass	fail	unavailable	fail	56
TACE-PADDING padding supervision	fault	pass	fail	unavailable	fail	56
TACE-LAST-POSITION final position supervision	fault	pass	fail	unavailable	fail	56
TACE-BENIGN-MIN equivalent mask expression	conformant	pass	fail	unavailable	pass	56
TACE-BENIGN-RENAME local variable rename	conformant	pass	fail	unavailable	pass	56
TACE-MISSING-EVIDENCE evidence eligibility	not evaluated	not evaluated	not evaluated	unavailable	not evaluated	0
TACE-ZERO-ELIGIBLE evidence eligibility	not evaluated	not evaluated	not evaluated	unavailable	not evaluated	0
AABS-v2-paired_conjunction-original_conformant paired conjunction	conformant	pass	unavailable	unavailable	pass	4
AABS-v2-paired_conjunction-fault paired conjunction	fault	pass	unavailable	unavailable	fail	4
AABS-v2-paired_conjunction-path_relevant_benign paired conjunction	conformant	pass	unavailable	unavailable	pass	4
AABS-v2-paired_conjunction-missing_source_file paired conjunction	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-paired_conjunction-actual_ineligible_input paired conjunction	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-operational_combined-original_conformant operational combined	conformant	pass	unavailable	unavailable	pass	4
AABS-v2-operational_combined-fault operational combined	fault	pass	unavailable	unavailable	fail	4
AABS-v2-operational_combined-path_relevant_benign operational combined	conformant	pass	unavailable	unavailable	pass	4
AABS-v2-operational_combined-missing_source_file operational combined	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0

Case	Expected	Field/type	Original adapters	Native tests	Behavioral contract	Eligible
AABS-v2-operational_combined-actual_ineligible_input operational combined	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-conditional_denominator-original_conformant conditional denominator	conformant	pass	unavailable	unavailable	pass	2
AABS-v2-conditional_denominator-fault conditional denominator	fault	pass	unavailable	unavailable	fail	2
AABS-v2-conditional_denominator-path_relevant_benign conditional denominator	conformant	pass	unavailable	unavailable	pass	2
AABS-v2-conditional_denominator-missing_source_file conditional denominator	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-conditional_denominator-actual_ineligible_input conditional denominator	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-paired_marginal_product-original_conformant paired marginal product	conformant	pass	unavailable	unavailable	pass	2
AABS-v2-paired_marginal_product-fault paired marginal product	fault	pass	unavailable	unavailable	fail	2
AABS-v2-paired_marginal_product-path_relevant_benign paired marginal product	conformant	pass	unavailable	unavailable	pass	2
AABS-v2-paired_marginal_product-missing_source_file paired marginal product	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-paired_marginal_product-actual_ineligible_input paired marginal product	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-informational_combined-original_conformant informational combined	conformant	pass	unavailable	unavailable	pass	1
AABS-v2-informational_combined-fault informational combined	fault	pass	unavailable	unavailable	fail	1
AABS-v2-informational_combined-path_relevant_benign informational combined	conformant	pass	unavailable	unavailable	pass	1
AABS-v2-informational_combined-missing_source_file informational combined	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-informational_combined-actual_ineligible_input informational combined	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-commit_success_flag-original_conformant commit success flag	conformant	pass	unavailable	unavailable	pass	1
AABS-v2-commit_success_flag-fault commit success flag	fault	pass	unavailable	unavailable	fail	1
AABS-v2-commit_success_flag-path_relevant_benign commit success flag	conformant	pass	unavailable	unavailable	pass	1

Case	Expected	Field/type	Original adapters	Native tests	Behavioral contract	Eligible
AABS-v2-commit_success_flag-missing_source_file commit success flag	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-commit_success_flag-actual_ineligible_input commit success flag	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-ranking_columns-original_conformant ranking columns	conformant	pass	unavailable	unavailable	pass	4
AABS-v2-ranking_columns-fault ranking columns	fault	pass	unavailable	unavailable	fail	4
AABS-v2-ranking_columns-path_relevant_benign ranking columns	conformant	pass	unavailable	unavailable	pass	4
AABS-v2-ranking_columns-missing_source_file ranking columns	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-ranking_columns-actual_ineligible_input ranking columns	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-scenario_macro-original_conformant scenario macro	conformant	pass	unavailable	unavailable	pass	4
AABS-v2-scenario_macro-fault scenario macro	fault	pass	unavailable	unavailable	fail	4
AABS-v2-scenario_macro-path_relevant_benign scenario macro	conformant	pass	unavailable	unavailable	pass	4
AABS-v2-scenario_macro-missing_source_file scenario macro	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0
AABS-v2-scenario_macro-actual_ineligible_input scenario macro	not evaluated	not evaluated	unavailable	unavailable	not evaluated	0

Table 3: Nine additional implementation cases.

Case	Expected	Field/type	Original adapters	Native tests	Behavioral contract	Eligible
FRESH-APRM-REWARD-RANGE-reference terminal outcome range remapping	conformant	pass	fail	unavailable	pass	16
FRESH-APRM-REWARD-RANGE-fault terminal outcome range remapping	fault	pass	fail	unavailable	fail	16
FRESH-APRM-REWARD-RANGE-benign terminal outcome range remapping	conformant	pass	fail	unavailable	pass	16
FRESH-TACE-RESPONSE-ORDER-reference response token order corruption	conformant	pass	fail	unavailable	pass	12
FRESH-TACE-RESPONSE-ORDER-fault response token order corruption	fault	pass	fail	unavailable	fail	12

Case	Expected	Field/type	Original adapters	Native tests	Behavioral contract	Eligible
FRESH-TACE-RESPONSE-ORDER-benign response token order corruption	conformant	pass	fail	unavailable	pass	12
FRESH-AABS-SCENARIO-STATISTIC-reference outer macro mean to order statistic	conformant	pass	unavailable	unavailable	pass	8
FRESH-AABS-SCENARIO-STATISTIC-fault outer macro mean to order statistic	fault	pass	unavailable	unavailable	fail	8
FRESH-AABS-SCENARIO-STATISTIC-benign outer macro mean to order statistic	conformant	pass	unavailable	unavailable	pass	8

3 Mutation details

The four fixture validators detected all 30 selected mutations: eight equation changes, eight schema changes, eight logging changes, and six aggregation changes. The set includes duplicate temporal transformations. The separate eight-operator experiment detected seven mutations on its first execution. Its relational mutation survived because the input contained no equality case. Adding that case increased detection to eight of eight with the check unchanged.

Table 4: Eight externally motivated mutation operators, evaluated before and after adding an equality case to the input. The check remained unchanged.

Family	Check	Mutation	First run	After fixture correction
Arithmetic operator replacement	XM-01: AgentPRM target value	horizon - 1 - t → horizon - 1 + t	Detected	Detected
Relational operator replacement	XM-02: thresholded pass rate	reward ≥ threshold → reward > threshold	Missed (fixture lacked equality case)	Detected
Constant-value replacement	XM-03: AgentPRM terminal index	terminal offset 1 → 0	Detected	Detected
Unary operator replacement	XM-04: AgentPRM signed target	value → -value	Detected	Detected
Logical operator replacement	XM-05: AgentAbstain paired accuracy	act and abstain → act or abstain	Detected	Detected
Statement deletion	XM-06: ContractBench response logging	delete final response append	Detected	Detected
Statement deletion	XM-07: ToolACE required arguments	delete required-argument insertion	Detected	Detected
Logical operator replacement	XM-08: AgentAbstain commit scoring	not commit and judge → not commit or judge	Detected	Detected

4 Prospective AgentAbstain details

Table 5: Five AgentAbstain checks specified before execution.

Check	Exact source support	Implemented result	Classification	Alternative value
AABS-PAIR-01	README.md:23 (263 pairs; one should-act and one should-abstain task); README.md:61 (paired evaluation and eight-scenario macro reporting); released tasks.jsonl at locked dataset revision	526 rows; 263 pairs; 8 scenarios; 0 invalid pairs	conformant	
AABS-PAIR-02	AgentAbstain paper Appendix B.6, Eq. (3) (pair conjunction); README.md:61 (paired evaluation); eval/statistics/ analysis.py:258-260; four-cell exact fixture	paired accuracy 0.25	conformant	
AABS-CAR-01	AgentAbstain paper Appendix B.6, Eq. (4) (conditional definition); eval/statistics/ analysis.py:231-243 (explicit zero fallback); conditioned-denominator exact fixture	nonempty=0.50; empty=0	convention sensitivity	empty denominator represented as undefined / NA
AABS-COMMIT-01	AgentAbstain paper Section 2.2 and Appendix B.6; README.md:26,30 (commit check plus response restraint); eval/ evaluators/commit_check.py:12-15,44-49; eval/statistics/analysis.py:142-165; commit-by-response Cartesian fixture	only noncommit + passing judge is successful (1/4)	conformant	
AABS-MACRO-01	AgentAbstain paper Section 3.1 and Figure 3 (macro over eight scenarios); README.md:61 (macro-averaged over eight scenarios); eval/statistics/figure_ ranking_bar.py:103-120 (equal action-subtype weights inside scenario); unequal-action-type exact fixture	equal-subtype headline macro 0.750	convention sensitivity	pair-weighted scenario macro 0.625

The conditional accuracy definition leaves the empty denominator unspecified. The implementation returns zero; representing the result as undefined is another possible convention. On the aggregation input, equal subtype weighting gives 0.750 and pair weighting gives 0.625.

5 ToolACE conversion and sequence lengths

The earlier custom conversion retained 9,412 of 10,092 candidate targets (93.3%). The official conversion produces 9,964 examples: 9,394 with nonempty responses and 570 with empty responses supervised on EOS only. Conversation-schema validation excludes 128 candidates. The official and custom converters disagree on 18 retained prompt/response rows; the conversion comparison records these differences.

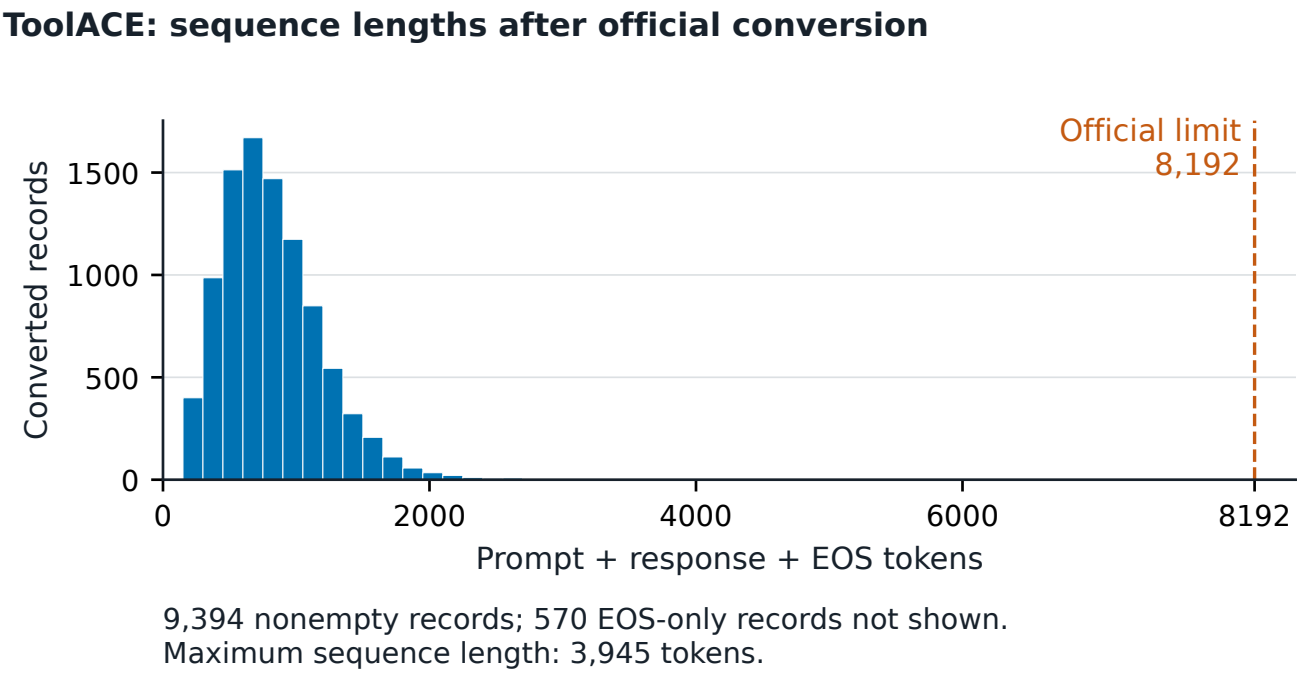


Figure 1: Sequence lengths for the 9,394 converted examples with nonempty responses. The 570 EOS-only examples are omitted from the histogram. All 9,964 converted examples were checked against the 8,192-token limit; none exceeded it, and the maximum length was 3,945 tokens.

6 Missing-pair analysis

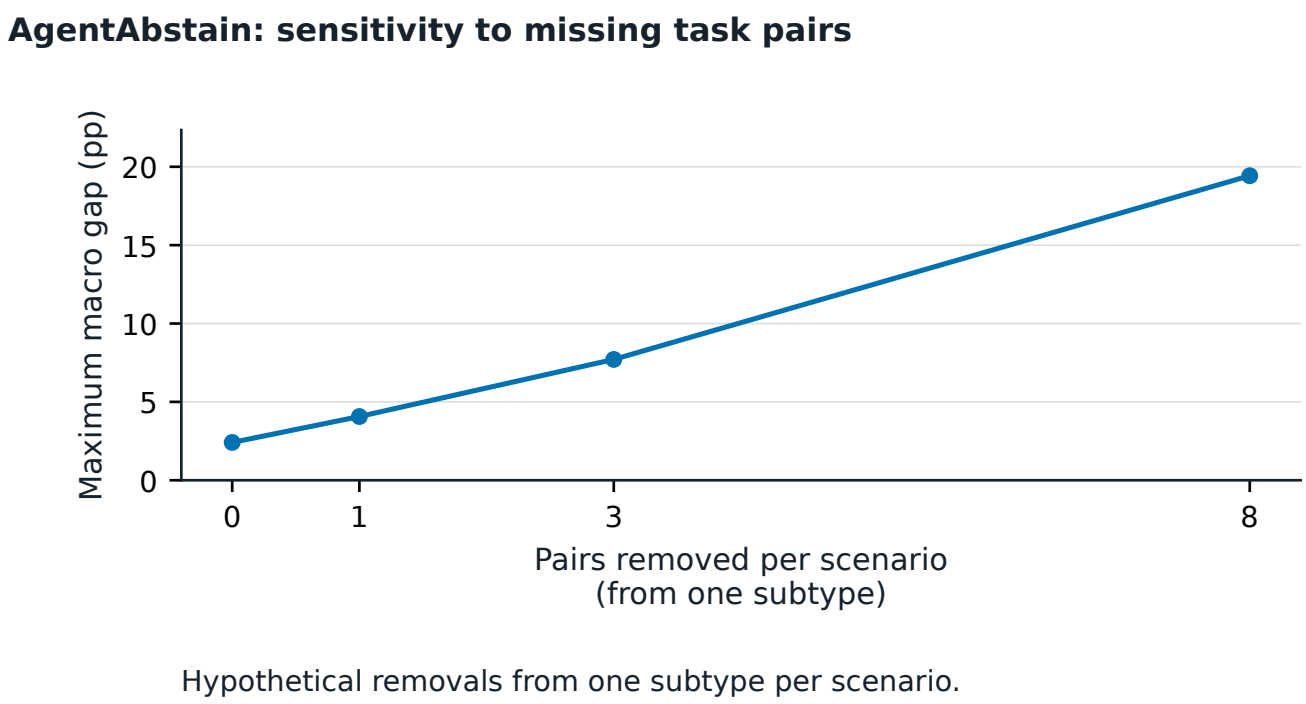


Figure 2: Maximum weighting difference after hypothetical removal of zero, one, three, or eight complete pairs from one subtype in each scenario. The removed subtype is chosen separately for each scenario to maximize the difference. Removing pairs simultaneously across the eight scenarios gives overall bounds of approximately 2.41, 4.07, 7.71, and 19.43 percentage points.

7 ContractBench comparison

Item	Evidence
Reported result	Qwen3.5-9B, n=99, 56.6%
Released artifact	Qwen3.5-9B, n=99, 0/99 success labels
Matches	model name; denominator
Not verified	checkpoint; task set; evaluation configuration
Trace issue	3/99 traces report 8,192 output tokens but contain no terminal assistant response
Interpretation	0/99 successes under the repository mapping; correspondence to the reported evaluation remains unresolved

Table 6: Reported ContractBench result and released evaluation records for Qwen3.5-9B.

8 Comparison with conventional tests

Field/type validation accepts AgentPRM's query-history error because observations and actions have the same string type. ContractBench traces can also retain expected field types while omitting a terminal response. ContractBench's existing regression test covers terminal-response logging. The implementation comparison in this paper evaluates the selected AgentPRM, ToolACE, and AgentAbstain behaviors.

9 Execution time after contract authoring

Repository	Median (s)	IQR (s)
AgentPRM	0.098	0.096–0.100
ContractBench	0.115	0.113–0.117
ToolACE	0.102	0.100–0.103
tau-bench	0.233	0.232–0.235
AgentAbstain	0.393	0.385–0.402

Table 7: Execution time of the original contract invocations. Median and interquartile range from seven runs per repository on an Apple Silicon CPU, including process startup.

10 Repository revisions

We evaluated AgentPRM at `e4714717`, ContractBench at `c50eefee`, ToolACE at `1156d649`, τ-bench at `59a200c6`, and AgentAbstain at `f5812497`. The AgentAbstain dataset revision is `84222842`. Full commit hashes and the dataset SHA-256 are included in the repository.